

Dynamic Coupling of Lunar Capture and Martian Migration: Evidence of a Planet 9 Flyby in the Inner Solar System

Alexssandro N. de Oliveira

Paradigm Zero Project

alexwztrong@gmail.com

March 9, 2026

Abstract

The current orbital architecture of the inner Solar System presents dynamic and angular momentum anomalies that challenge classical models of slow accretion and isolated evolution. This paper proposes a unified, purely mechanical N-body dynamical model that couples three fundamental geophysical and orbital events: Earth's axial tilt (23.5°), the insertion of the Moon into its current orbit, and the abrupt migration of Mars to 1.52 AU. Using high-precision numerical integrations (REBOUND IAS15 / RK4) under a strict methodological constraint of "Zero Modulation" (utilizing only proven original masses and mean motion physics), we demonstrate that the Earth-Mars system previously existed in a low-entropy state (3:2 resonance, zero terrestrial obliquity). The geometric rupture of this system is explained by the transient intrusion of a massive trans-Neptunian body (~ 44 Earth Masses), whose orbital characteristics—blindly demanded by our genetic algorithms—mathematically coincide with the theoretical constraints of the hypothetical Planet 9.

1. Introduction: The Epistemological Diagnosis

The orbital mechanics of the inner Solar System are conventionally interpreted through the lens of slow and continuous dissipation processes. However, the orbital, chemical, and geophysical signatures of the Earth-Moon-Mars system hold scars of recent dynamic events that require a strict mechanical review based purely on N-body interactions.

[IDENTIFIED DOMINANT PARADIGM]: The standard academic model assumes that the architecture of the Earth-Moon system was forged by a stochastic collision event (The Giant Impact Hypothesis - *Theia*), which is directly responsible for Earth's axial tilt ($\sim 23.5^\circ$) and the creation of the Moon from molten debris. In parallel, the current position of Mars (1.52 AU) is justified by passive orbital migration (The *Grand Tack* Hypothesis).

[LIMITATION OF THIS PARADIGM]: The Giant Impact Hypothesis structurally fails when attempting to reconcile the mechanics of an oblique collision with the **perfect isotopic homogeneity** (oxygen and titanium signatures) observed between Earth and the Moon. To

function, current models require improbable premises of total vaporization and mixing. Furthermore, Mars's orbit presents mass and positional anomalies indicating an abrupt dynamic ejection, not a passive accommodation. The central flaw of the current model is treating the geological and orbital configuration of these three bodies as temporally disconnected phenomena.

[APPLIED ALTERNATIVE PARADIGM]: The current chaotic configuration is, in fact, the scar of a violent N-body rupture that occurred in a system that previously possessed perfectly stabilized primary mechanics. The mechanical model is divided into the following formative phases:

1. **Co-accretion (The Isotopic Origin):** Earth and the Moon initially formed as co-orbital bodies ("sisters") in the same primary accretion cloud, which directly answers for the perfect isotopic identity between both, without any need for destructive impacts.
2. **Martian Separation and Low Entropy:** The entry of Mars into the forming inner system and its search for resonance with Earth forced the natural expulsion of the Moon into an external solitary orbit, where it cooled and consolidated its surface characteristics. Without the Moon, Earth and Mars locked into a coupled, low-entropy mechanical state (symmetrical resonant periods, non-inclined axes).

The throwing of the Moon back into Earth's orbit (generating the 23.5° torque on Earth) and the simultaneous ejection of Mars to 1.52 AU are not independent formation processes, but rather a single chain-reaction mechanical event triggered by a massive external perturber.

2. The Low Entropy Hypothesis and the Original State (Phase T-1)

To mathematically reconstruct the mechanics of the original Earth-Mars system (after the primary expulsion of the Moon), we established the rule of **"Zero Modulation"**. No orbital magnitude was biasedly arbitrated to force a preferred result. The *inputs* strictly utilized masses established by the IAU and proven original geological data (such as the reincorporation of Mars's recent global water mass, estimated at 2.3×10^{19} kg, into its current rocky mass).

The adopted principle is based on the observable mechanics of closed rocky planetary systems (such as

TRAPPIST-1), which naturally converge to mean motion resonances and zero axial inclinations.

2.1. Earth-Mars Stability Scan (Simulation T-1)

To find the exact coordinates of Mars’s orbit on the eve of the lunar capture event, we processed a grid of **864 N-body scenarios**.

The numerical integrations utilized the adaptive symplectic integrator **IAS15** (*REBOUND*), scanning first-order resonances (4:3, 3:2, 2:1, and 3:1). The analytical physics considered the shifts in Earth’s Center of Mass (CM) due to ocean distribution, and Mars’s CM based on its hemispheric dichotomy and hydrological network.

Analytical Results (The Winning Scenario 488)

The physics engine unambiguously converged to a state of absolute dynamic stability (Dynamic Score = 1.000 / var_a = 0.000002), revealing the following architecture for the original system:

- **Earth:** Orbited in a circular resonance with an exact period of 360 days. **Its native axial tilt was 0°**, receiving stellar radiation isometrically (absence of the extreme meteorological seasons observed today).
- **Mars:** Operated locked in a 3:2 gravitational resonance with Earth, possessing a period of exactly 540 days and orbiting at **1.3572 AU** (distant from its current orbit of 1.52 AU).

The simulation proves that, before the rupture event, the inner system was a perfectly stabilized thermodynamic and mathematical gear. The Moon orbited solitarily, awaiting the throwing event that would make it cross the terrestrial zone, while Mars operated as a stabilizing pendulum at 1.35 AU.

3. Lunar Injection and the Martian Migration Paradox

Having established the original state of thermodynamic rest (Earth at 0° / 360 days; Mars at 1.35 AU / 540 days), the rupture event of the inner system manifests through the anomalous intrusion of the Moon. Unlike slow *in situ* formation models, we treat the Moon’s arrival at its current orbit as a recent and **violent dynamic capture event**.

3.1. Dynamic Lunar Capture and Axial Torque (RK4 Integration)

To test the mechanical viability of an external lunar insertion into Earth’s sphere of influence, we developed a numerical integration engine using the 4th-Order Runge-Kutta method (RK4) optimized for N-bodies (*Phase_1_Final_Code_NBody.py*). The algorithm simulated the approach of the Moon—previously isolated in

the system—crossing Earth’s Hill Sphere on a hyperbolic trajectory.

Capture Integration Results:

- **Braking Time:** The gravitational capture process (initial circularization) occurred in an extremely short window of **~6.5 terrestrial days**.
- **Insertion Radius:** The Moon stabilized its initial orbit at a distance of approximately **60.59 Earth Radii ($R_{\oplus}$)**.
- **Angular Momentum Transfer:** The abrupt braking of a 7.34×10^{22} kg body in just 6.5 days generated a massive transfer of angular momentum to the Earth system. This gravitational shock provided the exact mechanical torque necessary to tilt Earth’s axis of rotation from its native 0° inclination to the current **23.5°**, destroying the planet’s climatic isometry and instituting the current extreme seasons.

3.2. Absolute Chronology: Tidal Regression and the 12,000-Year Window

The Dominant Paradigm fails to date the origin of the Earth-Moon system when attempting to reconcile the current rate of lunar recession with the geological age of the Earth (4.5 billion years). Through *Lunar Laser Ranging* (Apollo) measurements, it is known that the Moon recedes from Earth at a continuous rate of **3.8 cm/year**.

If this rate is linearly extrapolated into deep time (as classically demonstrated by Kaula’s tidal equations), the Moon would have been colliding with Earth or melting the terrestrial crust just 1.5 billion years ago—a fatal geological contradiction, frequently bypassed by academia with arbitrary continental resonance variables.

The **Paradigm Zero** resolves this contradiction by removing the premise of deep time.

We developed a reverse tidal validation algorithm (*Phase_2_Reverse_Validation_Code_Moon.py*). By applying the strict mathematical regression of the 3.8 cm/year rate, starting from the current position back to the exact insertion radius obtained in our RK4 simulation ($\sim 60 R_{\oplus}$), the algorithm returns an unequivocal timestamp: the lunar capture occurred exactly **12,077 years ago**.

This 12-millennium window perfectly aligns with the Pleistocene-Holocene boundary, a period marked by geological cataclysms, abrupt mass extinctions, and the end of the planet’s previous thermodynamic state. The Moon has not orbited Earth for billions of years; it is a recent intruder.

3.3. The Mars Paradox (The Gravitational Limit of the Isolated System)

The injection of the Moon 12,000 years ago perfectly explains Earth’s axial anomaly. However, the N-body sys-

tem reveals an intractable mechanical bottleneck if we consider the Solar System as an isolated environment.

When running the lunar capture simulation while keeping Mars in its original orbit of 1.35 AU (`05_Phase_1_Final_Output.txt`), we observed the following behavior in the integrator’s *logs*:

1. The Moon invades Earth’s orbit.
2. Earth absorbs the angular momentum and tilts its axis.
3. **Mars remains frozen in its orbit of 1.3572 AU.**

The Paradox Realization: The combined mass of the Moon (7.3×10^{22} kg) and Earth (5.9×10^{24} kg) is gravitationally insufficient to eject Mars (6.39×10^{23} kg, plus its global paleo-ocean) from 1.35 AU to its dead orbital position of 1.52 AU.

If the Moon caused the terrestrial cataclysm, what caused the ejection of Mars and the injection of the Moon itself?

The law of conservation of energy and momentum dictates that the amount of kinetic energy required to **simultaneously** (1) throw the Moon from its external orbit against Earth and (2) abruptly eject Mars from 1.35 AU to 1.52 AU, could only be provided by a **Massive External Agent**. This body needed to cross the inner Solar System in the exact same geological timeframe ($\sim 12,000$ years).

This mathematical necessity forces us to seek the gravitational signatures of a trans-Neptunian perturber on a high-eccentricity trajectory: the so-called Planet 9.

4. Genetic Optimization of the External Agent: The Signature of Planet 9

Faced with the mechanical impossibility of the Earth-Moon system ejecting Mars from its primary low-entropy orbit (1.3572 AU) to its current orbit (1.524 AU), the premise was established that the lunar capture event $\sim 12,000$ years ago was collateral to the passage of a massive intrusive body.

To determine the physical and orbital characteristics of this mysterious perturber without bias, we avoided anchoring the simulation in pre-existing theories. We employed a rigorous **Genetic Algorithm (GA)** coupled with an N-body engine (`Planet_9_GA_RK4_Complete.py`).

The objective of the algorithm’s *fitness function* was singular: to find the lowest possible margins of error capable of displacing Mars to exactly 1.52 AU, injecting the Moon into Earth’s orbit, and preserving Earth stabilized at 1.00 AU.

4.1. Convergent Parameters of the Perturber (Phases 3 and 4)

After successive generations of evolutionary optimization, processing thousands of mass variations and ap-

proach vectors, the algorithm mathematically converged to a highly specific mass and trajectory profile.

Any body with lower mass would fail to displace Mars sufficiently; any more massive body, or one with closer proximity, would eject Earth from the Solar System or cause it to collide with Venus.

Genetic Algorithm Results (Generation 50 / Winning Individual):

- **Required Mass:** ≈ 44 Earth Masses ($M_{\oplus}$).
- **Interception Velocity** (v_{inf}): ≈ 44.8 km/s.
- **Perihelion Distance of Passage** (q): An extreme grazing pass at only **0.0086 AU** from Earth’s orbit.
- **Trajectory Inclination:** $\approx 17.5^\circ$ relative to the ecliptic plane.
- **Passage Eccentricity** (e): 1.05 (Escape/hyperbolic trajectory at the moment of the *flyby*).

The Planet 9 Signature: The mass of $44 M_{\oplus}$ demanded by our independent calculations to solve the "Mars Paradox" remarkably aligns with the upper theoretical constraints proposed by contemporary astrophysics for the hypothetical **Planet 9** (or *Planet X*).

Classical models based on the anomalous clustering of Kuiper Belt Objects (KBOs) suggest the existence of an ice giant or super-Earth in a highly elliptical orbit at the edges of the Solar System. Our dynamic simulations of the inner system demand the existence of this exact body, proving that its elliptical orbit brings it, periodically or in a singular past event, into the habitable zone.

4.2. The Dynamics of the Rupture: The End of Low Entropy

With the parameters of the External Agent defined ($44 M_{\oplus}$, 0.0086 AU), we executed the chronological simulation of the transverse passage (`Simulation_Flyby_NBody.py`). The sequence of mechanical events that destroyed the 3:2 resonance (Earth-Mars) decodes the current state of the inner Solar System:

1. **The Lunar Injection (The Trigger):** The External Agent, while crossing the Solar System plane at ~ 44.8 km/s, gravitationally perturbed the Moon (which was orbiting isolated after its primordial expulsion). The perturber’s gravitational vector forced the Moon into a hyperbolic intersection trajectory with Earth, resulting in rapid capture (~ 6.5 days) and the tilting of Earth’s axis to 23.5° (Section 3.1).
2. **The Martian Ejection (Breaking the Resonance):** Immediately after (or simultaneously with) the lunar perturbation, the $44 M_{\oplus}$ mass of the External Agent, passing at a mere 0.0086 AU, overcame the stability of the Martian gravitational pendulum.

The combined tidal force ripped Mars from its circular orbit at 1.3572 AU.

3. **The Loss of the Martian Ocean:** The abrupt acceleration and deformation of the gravitational potential well forced the catastrophic ejection of a large part of the primitive atmosphere and the recent global Martian ocean (the previously calculated 2.3×10^{19} kg). Mars was slingshotted to the dead, elliptical orbit of 1.524 AU, where it froze in its current state.
4. **Terrestrial Survival:** Earth, anchored by its greater mass (5.9×10^{24} kg) and now stabilized by the new, massive angular momentum of the newly formed binary system with the Moon, resisted ejection, maintaining itself at 1.00 AU, albeit inheriting the climatic chaos of its newly inclined axis.

This chain of events, numerically validated, concludes the rupture model. However, it raises a question of systemic momentum conservation: if Planet 9 executed this devastating passage $\sim 12,000$ years ago in a hyperbolic orbit ($e = 1.05$), how was it originally captured, or conversely, why did it not return with the same violence?

To answer this, the simulations require the inclusion of the planetary system's largest gravitational well: the barycenter of Jupiter.

5. Dynamic Dispersion of the Perturber: Coupling with Jupiter

The N-body rupture model of the inner Solar System, validated by the Genetic Algorithm, determined that the External Agent (Planet 9, with $44 M_{\oplus}$) crossed the terrestrial zone with an eccentricity of $e \approx 1.05$. This is a locally hyperbolic trajectory, which suggests an escape vector.

For this body not to be ejected from the Solar System and to consolidate into the current theoretical trans-Neptunian orbit estimated by academia (semi-major axis of $\sim 400 - 800$ AU), the massive angular momentum of this passage needed to be internally dissipated. The law of conservation of energy requires the intervention of a gravitational well capable of absorbing and braking the kinetic energy of $44 M_{\oplus}$ moving at 44.8 km/s.

To validate this dissipation mechanics, we expanded the N-body simulation (`Simulation_Flyby_With_Jupiter_NBody.py`) to include the barycenter of Jupiter (1.898×10^{27} kg) as the angular modulator of the system.

5.1. The "Jupiter Shield" and Trans-Neptunian Orbital Stabilization

The simulations processed the exit trajectory of the External Agent post-event at Earth (after the throwing of

the Moon and Mars to 1.52 AU). The results unequivocally demonstrated that the intrusive body did not escape into the interstellar medium but was captured in a high-energy interaction with Jupiter.

The Mechanics of the Jupiter-Planet 9 Coupling:

1. **Approach and Braking:** After devastating the inner system, the External Agent crossed Jupiter's orbit. The immense gravitational influence of the gas giant forced a sharp deflection curve in the intruder's trajectory.
2. **Eccentricity Conversion:** Jupiter acted as a massive orbital brake, converting the locally hyperbolic orbit ($e \approx 1.05$) into a closed, albeit extremely elongated, elliptical orbit ($e \approx 0.90$ to 0.98).
3. **Stabilization Window:** The time measured by the simulation from the passage through terrestrial perihelion (0.0086 AU) to the final crossing and braking by Jupiter's orbit lasted approximately **500 terrestrial years**. During this half-millennium period, the perturber remained within the inner Solar System, acting as a secondary destabilizing agent before being slingshotted toward the Oort Cloud.

This interaction with Jupiter corroborates not only the survival of the Solar System itself but solves the anomaly of "**Planet 9**". The mathematics proves that the anomalous mass indirectly detected at the edges of the system by contemporary astronomers (based on the perturbation of KBOs) is the exact dynamic gravitational fossil of the $44 M_{\oplus}$ body that reconfigured Earth and Mars 12,000 years ago.

6. Conclusions and Implications of the Paradigm Zero

The present study introduces a unified restructuring of the geodynamic history of the inner Solar System, replacing the postulate of gradual isolated evolution and formative stochastic impacts (*Theia* / *Grand Tack* models) with a singular and catastrophic N-body event, mathematically pinpointed to $\sim 12,000$ years ago.

Through the strict application of REBOUND IAS15 integrations, Genetic Algorithms (RK4), and restricted gravitational regression, the **Paradigm Zero** demonstrates that:

1. **Co-Orbital Origin and Isotopic Identity:** Earth and the Moon formed as sister bodies in the same accretion cloud (explaining the perfect isotopic identity). The primordial consolidation of the system and the early migration of Mars isolated the Moon in a cold, external orbit, separating their formative histories.
2. **T-1 Low Entropy State:** The system operated in continuous thermodynamic equilibrium. Earth orbited with a null axis (0°), free of severe seasons,

in a 360-day year. Mars orbited in a 3:2 resonance, stabilized at 1.3572 AU.

3. **The Failure of Theia and Recent Lunar Capture:** The current 23.5° inclination of Earth’s axis is not an artifact of primordial formation but the mechanical record of the violent capture of the Moon, which occurred in just ~ 6.5 days. The rigorous regression of the current tidal rate of 3.8 cm/year anchors this event precisely 12,077 terrestrial years ago (coinciding with the Holocene boundary).
4. **The Rupture of the Mars Paradox by Planet 9:** The force required to eject Mars, its oceans, and its atmosphere from the stable position at 1.35 AU to the eccentric desolation at 1.52 AU exceeds the angular capacity of the Earth-Moon system. The intervention of a 44 Earth Mass body crossing at 0.0086 AU was required. This External Agent, subsequently contained by Jupiter’s gravity in a 500-year process, currently inhabits the outer Solar System, being the theoretical entity responsible for the Kuiper anomalies (Planet 9).

The **Paradigm Zero** reconciles N-body computational astrophysics with the signatures of climatic cataclysms and mass extinctions attested in the terrestrial crust at the exact boundary of the **Pleistocene-Holocene** ($\sim 12,000$ years ago), in addition to deciphering the mystery of a dead Mars. It replaces a model of peaceful coalescence that requires billions of years with brutal, documentable thermodynamic dynamics.

The next stage of verifying this hypothesis lies in the observational localization of Planet 9, whose mass and escape vector have been numerically delimited in this work.

Data Availability and Open-Source Code

In adherence to the principles of strict scientific transparency and the “Zero Modulation” methodology, all raw data, N-body simulation source codes (Python/REBOUND IAS15), Genetic Algorithm parameters, and exact integration logs utilized in this study are open-source. They are fully available for peer replication and independent verification at the official GitHub repository:

[https://github.com/alewztrong/
paradigm-zero-simulations](https://github.com/alewztrong/paradigm-zero-simulations)

Researchers are encouraged to execute the codes sequentially (Phases T-1 to 5) to observe the dynamic coupling of the lunar capture, the Martian migration, and the Planet 9 flyby exactly as described in this manuscript. The methodological preservation of the 1.29 margin of error is fully documented within the repository to ensure computational authenticity.

Bibliographic References

- **REBOUND:** An open-source multi-purpose N-body code for collisional dynamics (IAS15 Integrator).
- **Runge-Kutta Methods:** For Orbital Mechanics (RK4 implementation applied to Genetic Algorithms).
- **Lunar Laser Ranging:** Secular Acceleration of the Moon (Kaula’s equations adapted for strict N-body regression).